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Integer linear programming problem (ILPP)

Pure ILPP

All decision variables are restricted to integers

$x_1, x_2, x_3 \geq 0$ and integers

Mixed ILPP

Some of decision variables are restricted to integers

$x_1, x_2, x_3 \geq 0$ and x_2 is an integer

Methods

1. Branch and bound method (Graphically) (Both Pure ILPP and Mixed ILPP)
2. Gomory's Cutting Plane Method
   - Case 1 → Pure ILPP
   - Case 2 → Mixed ILPP

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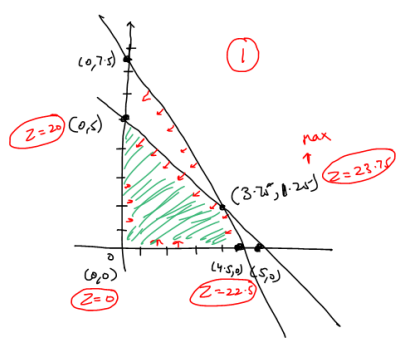
Method 1 Branch and bound Method

Q Max $Z = 5x_1 + 4x_2$

s.t. $x_1 + x_2 \leq 5$

$10x_1 + 6x_2 \leq 45$

$x_1, x_2 \geq 0$ and integers



Reduced LPP

Max $Z = 5x_1 + 4x_2$

s.t. $x_1 + x_2 \leq 5$

$10x_1 + 6x_2 \leq 45$

$x_1, x_2 \geq 0$

Step 1: Find the optimal solution (graphically) of Reduced LPP

$x_1 = 3.75$, $x_2 = 1.25$ and $Z = 23.75$

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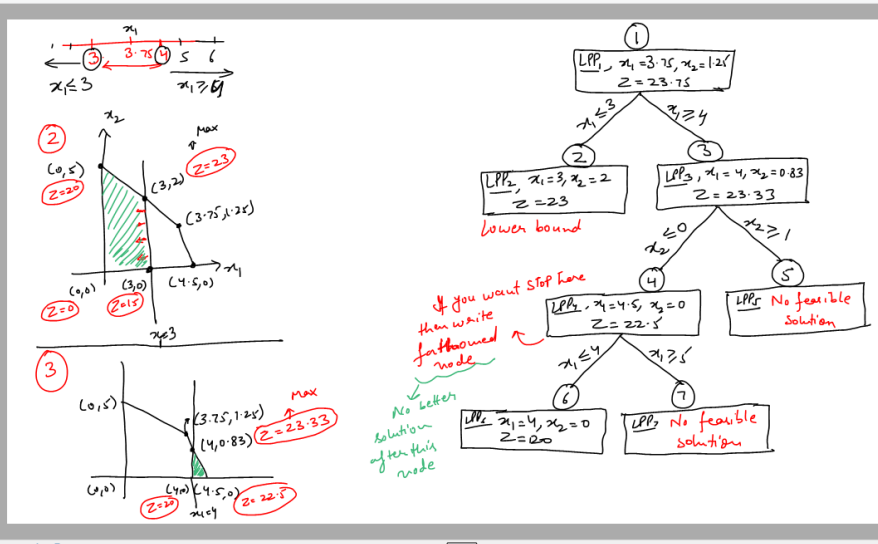
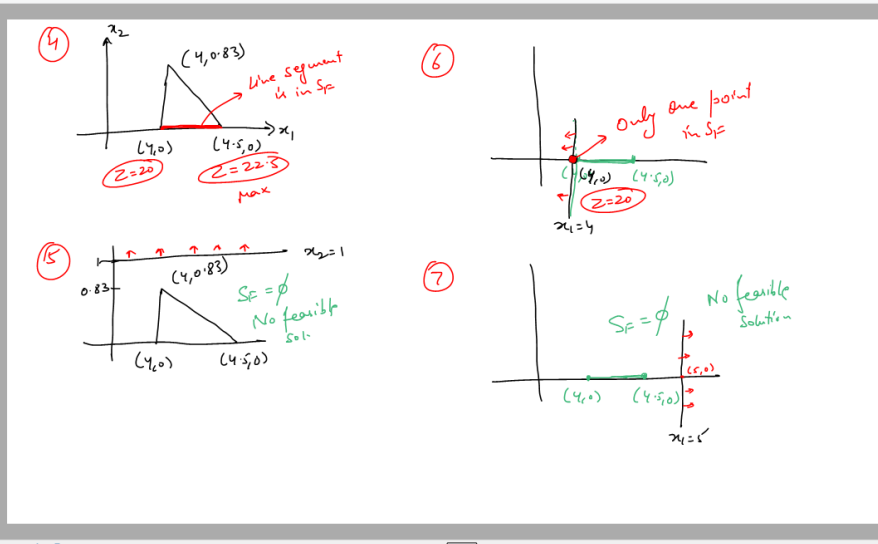

$x_1 \le 3$
$x_1 \ge 4$
(2) x_2 Max $Z=23$ (0,5) $Z=20$ (3,2) (3.75,1.25) (0,0) $Z=0$ (3,0) $Z=15$ (4.5,0) x_1 $x_1=3$
(3) (0,5) (3.75,1.25) (4,0.83) Max $Z=23.33$ (0,0) (4,0) $Z=20$ (4.5,0) $Z=22.5$ $x_1=4$
(1) LPP1, $x_1=3.75$, $x_2=1.25$, $Z=23.75$
$x_1 \le 3$ $x_1 \ge 4$
(2) LPP2, $x_1=3$, $x_2=2$, $Z=23$ lower bound
(3) LPP3, $x_1=4$, $x_2=0.83$, $Z=23.33$
$x_2 \le 0$ $x_2 \ge 1$
(4) LPP4, $x_1=4.5$, $x_2=0$, $Z=22.5$
(5) LPP5 No feasible solution
If you want STOP here then write fathomed node
No better solution after this node
$x_1 \le 4$ $x_1 \ge 5$
(6) LPP6 $x_1=4$, $x_2=0$, $Z=20$
(7) LPP7 No feasible solution


(4) x_2 (4,0.83) Line segment is in S_F x_1 (4,0) (4.5,0) $Z=20$ $Z=22.5$ Max
(5) $x_2=1$ 0.83 (4,0.83) $S_F=\phi$ No feasible sol. (4,0) (4.5,0)
(6) Only one point in S_F (4,0) (4.5,0) $Z=20$ $x_1=4$
(7) $S_F=\phi$ No feasible solution (5,0) (4,0) (4.5,0) $x_1=5$

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Q Min $Z = 5x_1 + 4x_2$

s.t. $3x_1 + 2x_2 \geq 5$

$2x_1 + 3x_2 \geq 7$

$x_1, x_2 \geq 0$ and integers

Reduced LPP Min $Z = 5x_1 + 4x_2$

s.t. $3x_1 + 2x_2 \geq 5$

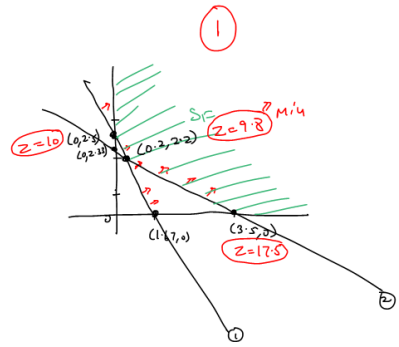
$2x_1 + 3x_2 \geq 7$

$x_1, x_2 \geq 0$

Solution of Reduced LPP is

$x_1 = 0.2, \quad x_2 = 2.2$

Min $Z = 9.8$



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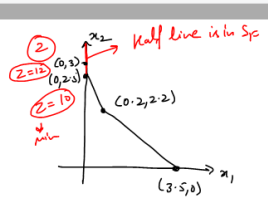

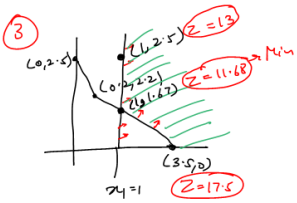

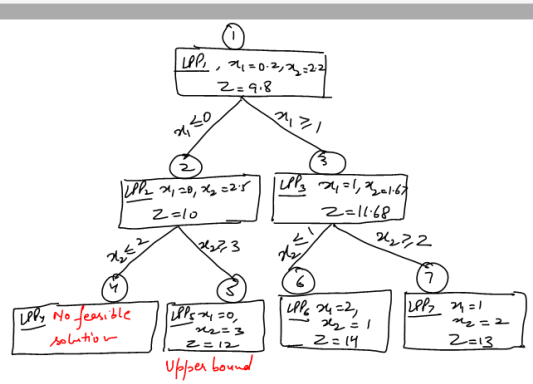

optimal Solution, $x_1 = 0$, $x_2 = 3$

with Min $Z = 12$

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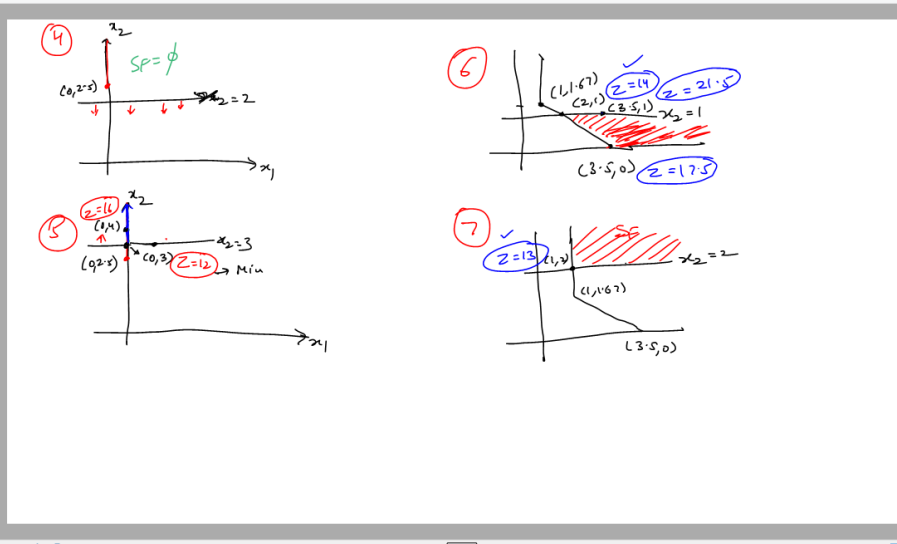

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Method 2 Gomory's Cutting plane Method

Case 1 : Pure ILPP

Optimal Table of Reduced LPP

B.V	x_1	x_2	s_1	s_2	Sol.
Z	0	4/3	0	1/6	3/2
s_1	0	5/3	1	-1/6	5/2
x_1	1	1/3	0	1/6	3/2

T_1

Q Max $Z = x_1 - x_2$

s.t. $x_1 + 2x_2 \le 4$

$6x_1 + 2x_2 \le 9$

$x_1, x_2 \ge 0$ and integers

Let us consider x_1-row from T_1,

x_1-row: $\boxed{x_1 + \frac{1}{3}x_2 + \frac{1}{6}s_2 = \frac{3}{2}}$

In General, $x_1 + a_1y_1 + a_2y_2 + \cdots + a_py_p = a$

(where $a_1 = \frac{1}{3}$, $y_1 = x_2$
$a_2 = \frac{1}{6}$, $y_2 = s_2$
$a = \frac{3}{2}$)

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